PORTAL Trial

Clinical formulas used in the CRFs

Field name, stored key, and the equation | 12 November 2026

This list is the calculated fields only — values the screen works out, not every typed number. Web and tablet use the same equations. Tablet Helper 3 is web Helper 4 (Infect / GI / Hema). Tablet Helper 4 is web Helper 5 (Metab / Renal / Eye). Database keys are not renamed.

Weight in formulas is always kilograms (birth_weight grams ÷ 1000, or DMS 5.7.A grams ÷ 1000). Day of life on scans is 1-indexed (birth calendar day = DOL 1). Forms AE, Y, SAE listing and Form L have no clinical calculators.

Form A — Screening

Nurses enter best gestational age. EDD is still derived from LMP when LMP is present. Same on web and tablet.

Screen field	Stored / API field	Formula
2 (b). EDD	expected_delivery_date	Naegele: EDD = LMP + 1 year − 3 months + 7 days. Not LMP + 280 days (that is 1–2 days early when the 9-month span has extra 31-day months).
LMP from EDD (if only EDD is known)	lmp_date	Inverse Naegele: LMP = EDD − 1 year + 3 months − 7 days.
1 (a) / 1 (b). Best estimate of GA	gestation_weeks, gestation_days	If checking against LMP: completed weeks = floor(calendar days from LMP to screening date / 7); days = remainder. Direct entry of weeks+days is the live path.
Eligibility window	—	Best GA must be 25+0 to 31+6 weeks. Screening ID is assigned only inside this window.

Form B — Birth and resuscitation

Weight is stored in grams. Centile uses Male/Female only; if sex is DSD the chart uses designated gender.

Screen field	Stored / API field	Formula
Gestation at randomisation (weeks + days)	gestation_rand_weeks, gestation_rand_days	Screening GA in days (weeks × 7 + days) + calendar days from screening date to date of birth. Then weeks = floor(total / 7), days = remainder. Never negative.
16. Intrauterine growth status (centile)	intrauterine_centile	INTERGROWTH-21st Very Preterm birthweight tables (Villar et al., Lancet 2016), 24+0 to 32+6 weeks, Male/Female. Birth weight kg is compared with the published 3rd, 5th, 10th, 50th, 90th, 95th, 97th kg for that exact week+day (no interpolation). Stores the lower published point (3, 5, 10, 50, 90, 95, 97) or "<3rd centile". Outside the chart, the nurse types it.
26. Strata (if randomised Yes)	strata	If GA at randomisation < 28+0 weeks → "< 28 weeks". Else → "≥ 28 – 31 weeks".
44. Cord clamp elapsed seconds	cord_clamp_time	Seconds from time of birth (clock) to cord-clamp clock, wrapping midnight. Allowed 0–300 seconds.
43. Cord clamp clock	cord_clamp_timestamp	Time of birth + elapsed seconds, wrapped to 24 hours. Auto-filled when elapsed seconds are typed.

Form C — Maternal details

Triple I does not use foul-smelling liquor. Steroid courses are counted from doses given.

Screen field	Stored / API field	Formula
Triple I	triple_i	If maternal fever = No → No. If fever = Yes: Yes when at least 2 of these 4 are Yes — fetal tachycardia, maternal TLC high, maternal tachycardia, abdominal tenderness. If all 4 are answered and fewer than 2 are Yes → No. Unresolved while those 4 are incomplete.
22. Antenatal steroid course status	derived from beta/dexa doses	Betamethasone: 1 course = 2 doses. Dexamethasone: 1 course = 4 doses. Status = Complete only when doses > 0 and doses divide evenly by doses-per-course; else Incomplete.
23. Number of completed courses	derived	floor(doses given / doses per course) for that drug.
MgSO4 gestation (weeks + days)	mgso4_gestation_weeks, mgso4_gestation_days	Preferred: GA at birth in days − (date of birth − MgSO4 date) in calendar days. If birth GA is missing: calendar days from LMP to MgSO4 date. Then weeks = floor(/7), days = remainder. Cleared if result is outside 0–314 days.
LMP / EDD (autofill from Form A)	lmp, edd	Same Naegele pair as Form A. Stored LMP/EDD first; missing one is completed from the other; if both missing, derived from screening GA + screening date.

Form D — Postnatal day 1

Caffeine loading is a single mg/kg dose. Maintenance is total mg/kg/day (not split by OD/BD).

Screen field	Stored / API field	Formula
3. Growth status	growth_status	From Form B intrauterine_centile (or recalculated INTERGROWTH if NBS GA is used): <10th centile → SGA; 10th–90th → AGA; >90th → LGA.
3 (a). SGA centile (if SGA)	sga_centile	"<3rd" if centile is "<3rd centile" or numeric < 3. "<10th" if numeric ≥ 3 and < 10.
16 (c). Caffeine loading dose (mg/kg)	display from caffeine_loading_abs; caffeine_dose also stored	mg/kg = absolute mg ÷ (birth_weight grams ÷ 1000). Typical 20 mg/kg. Soft warn if outside 10–25 mg/kg.
Typical loading amount (hint)	—	20 × birth weight in kg (mg).
16 (f). Caffeine maintenance (mg/kg/day)	display from caffeine_maint_abs	mg/kg/day = absolute mg ÷ birth weight kg. Soft warn if outside 5–10 mg/kg/day.
Effective GA for later forms	Form B gestation_weeks overlay / Form D NBS	If Form D method is NBS and |NBS days − Form B days| > 14, Forms F–K and helpers use the NBS GA. Form B itself keeps the original screening/birth GA.
13. Surfactant dose (mg/kg)	surfactant_dose	Entered as mg/kg (not calculated). Soft warn outside 50–200 mg/kg.

Form E — NICU admission

Age uses both date and time of birth and of admission.

Screen field	Stored / API field	Formula
Age at admission (hours)	age_at_admission_hours	floor((admission datetime − birth datetime) in minutes / 60). Empty if admission is before birth.
Age at admission (total minutes)	age_at_admission_minutes	floor((admission datetime − birth datetime) in milliseconds / 60 000). This is total minutes, not the remainder after hours. Displayed as "H hr M min" = floor(minutes/60) hr and minutes % 60.

Form F — Cranial USG

DOL is 1-indexed: the calendar day of birth is DOL 1. PMA adds completed postnatal days (DOL − 1).

Screen field	Stored / API field	Formula
Day of life (each scan)	dol	DOL = (scan date − date of birth) in calendar days + 1.
PMA (each scan)	pma_weeks / pma_days (shown as W+D wks)	PMA days = (GA weeks × 7 + GA days) + (DOL − 1). PMA weeks = floor(PMA days / 7); PMA days = remainder.

Form G — ROP screening

Same DOL/PMA arithmetic as Form F (birth day = DOL 1).

Screen field	Stored / API field	Formula
DOL at screening	dol	DOL = (screening date − date of birth) in days + 1.
PMA at screening	pma	PMA days = GA days at birth + (screening date − DOB). Shown as “Ww Dd”.
PMA at treatment / other event	derived	Same as screening PMA, using that event’s date instead of the screening date.

Form H — Neonatal morbidities

BPD is a suggestion only. The clinician applies Yes/No and grade on Form H. Assessment day is 36+0 weeks PMA.

Screen field	Stored / API field	Formula
36+0 PMA date (BPD assessment day)	—	Date = date of birth + (252 − GA days at birth) days. 252 = 36 weeks.
35–37. BPD / support / grade (Jensen 2019 suggestion)	bpd, bpd_support_36w, bpd_grade	From Helper 2 (or Form J 36-week visit if already left): room air → No BPD. NC ≤ 2 L/min → Grade 1. NC > 2 L/min, HFNC, CPAP or NIPPV → Grade 2. Intubated or SIMV / A-C / PSV / HFOV → Grade 3. Died before 36+0 → no suggestion. Still in NICU → Helper 2 that day (±1 if missing). Discharged/LAMA before 36+0 → Form J 36-week visit, else last Helper 2 day. Back-referred → Form J 36-week visit only.
PMA checkpoint dates (36 / 40 / 44 weeks) for Form I prefill	target_date	Date = Day 1 Date + (target PMA weeks × 7 − GA days at birth).

Form I — Study outcomes

One death is copied across Yes sections. Age uses times when both birth and death times are known.

Screen field	Stored / API field	Formula
Age at death — 7-day mortality	mortality_7d_age_hrs	Round((death − birth) in hours). Uses time of birth and time of death when both are present; otherwise dates only at 00:00.
Age at death — other mortality places	mortality_28d_age_days, death36_age_days, death40_age_days, death44_age_days, mortality_hospital_age_days, mortality_after_discharge_age_days	floor((death − birth) in hours / 24).
36 / 40 / 44 week PMA checkpoint date	target_date (prefill)	Same as Form H: Day 1 Date + (checkpoint weeks × 7 − GA days at birth).

Form J — External hospital outcomes

36-week visit respiratory fields also feed the Form H BPD suggestion.

Screen field	Stored / API field	Formula
Visit / assessment calendar date	from assessment_weeks	PMA target date = Day 1 Date + (assessment weeks × 7 − GA days at birth).
36-week BPD grading (via Form H suggestion)	resp_support, resp_mode, flow_rate	No support → No BPD. IMV → Grade 3. CPAP/NIPPV → Grade 2. Nasal cannula: Grade 1 if flow ≤ 2 L/min, Grade 2 if flow > 2.

Form K — MRI brain

PMA at MRI uses completed postnatal days (MRI date − DOB), not DOL+1.

Screen field	Stored / API field	Formula
PMA at MRI (weeks + days)	pma_weeks, pma_days	Postnatal days = MRI date − date of birth. PMA days = GA days at birth + postnatal days. Weeks = floor(/7), days = remainder. Empty if MRI is before birth.

Daily Monitoring Sheet (DMS)

Web and tablet. Slot frequency is chosen by the nurse (usually 2 h). Unlogged slots are not treated as NPO or as feeds.

Screen field	Stored / API field	Formula
GI total volume (ml)	summary (not a stored CRF field)	Sum of volume_ml on slots marked EF (enteral feed).
Volume by milk type	summary	Same sum, split by FM / EBM / PDHM.
EF hours / NPO hours	summary	Count of EF (or NPO) slots × feed-log frequency hours.
5.7.A Weight used downstream	growth_a.weight_g → kg	Latest weight on/before the helper calendar day, as kg = grams ÷ 1000. Helpers only switch from birth weight to this value once it is ≥ birth weight.
Day of life divider on the sheet	display	Day of life = floor((clock − birth datetime) / 24 h) + 1. The sheet shows when DOL ticks over on that calendar date.

Shared helper calendar (Helpers 2–5 and DMS)

Day 1 Date is date of birth. Before 08:00 local, the NICU "working day" is still yesterday.

Screen field	Stored / API field	Formula
NICU day number ("today")	nicu_day	Working date = calendar today, minus 1 day if local hour < 08:00. NICU day = (working date − Day 1 Date) in days + 1, minimum 1.
Calendar date for NICU day N	—	Day 1 Date + (N − 1) days.
Day of life (display on helpers)	—	Birth to <24 h = Day 1 of life; 24–48 h = Day 2; and so on. Helper tab numbers stay calendar-based (Day 1 = date of birth).

Helper 2 — Resp / CV / Neuro (tablet: Helper 2)

MAP vs CPAP which box to show is derived from the support modes. Values themselves are typed.

Screen field	Stored / API field	Formula
4. Max MAP or Max CPAP (which field)	map_cpap / map_cpap_secondary	From selected modes: NC/HFNC only → field N/A. CPAP only → CPAP. Pressure modes (NIPPV, SIMV, A/C, PSV, HFOV) → MAP. CPAP + a pressure mode → MAP plus 4 (b) CPAP.

Helper 3 — FiO2 logging (web Helper 3; not a tablet CRF calculator)

AUC uses fraction of inspired oxygen, not the percent number raw.

Screen field	Stored / API field	Formula
Row AUC	derived per row	AUC = (FiO2 percent ÷ 100) × duration hours.
Window AUC (each 12 h block)	derived	Sum of row AUCs in that window.
Day AUC	derived	Window 1 AUC + Window 2 AUC.
Grand total AUC	derived	Sum of day AUCs across logged days. Days with Helper 2 Supplemental O2 = Yes need both 12 h windows filled.

Helper 4 — Infect / GI / Hema (tablet: Helper 3)

Feed volume is ml/kg/day from cumulative millilitres, not a typed rate.

Screen field	Stored / API field	Formula
Effective weight (kg)	used for #15; not the nurse-typed field	Latest DMS 5.7.A weight (kg) on/before this NICU day, but only if it is ≥ birth weight kg. Until then (normal postnatal loss) use Form B birth_weight ÷ 1000. If only one of the two exists, use that one.
15. Feed volume (ml/kg/d)	feed_volume	cumulative_feed_volume (ml) ÷ effective weight kg, rounded to 1 decimal.

Helper 5 — Metab / Renal / Eye (tablet: Helper 4)

Nurses type millilitres per urine window. The rate and the daily total are ml/kg/hr. Same weight rule as Helper 4 feeds. PI 2026-11-11.

Screen field	Stored / API field	Formula
Effective weight (kg)	urine_effective_weight_kg	Same as Helper 4: DMS 5.7.A kg if recovered to/past birth kg, else birth weight kg.
8am–2pm urine rate (display)	from urine_output_8am_2pm (ml)	ml/kg/hr = ml ÷ (kg × 6 hours), 2 decimals.
2pm–8pm urine rate (display)	from urine_output_2pm_8pm (ml)	ml/kg/hr = ml ÷ (kg × 6 hours), 2 decimals.
8pm–8am urine rate (display)	from urine_output_8pm_8am (ml)	ml/kg/hr = ml ÷ (kg × 12 hours), 2 decimals.
Urine output total	urine_output_total	ml/kg/hr over recorded windows only: (sum of typed ml) ÷ (kg × hours of those windows). Blank or Not Recorded windows are skipped. Worked out on the server on save.
Window range (PI 2026-11-11)	—	A typed window > 15 ml/kg/hr is refused (HTTP 422) and will not save. Above 6 or below 0.5 ml/kg/hr is allowed but raises Data queries URINE_OUT_OF_RANGE (one per window). A blank window with no status, 24 h after the day, raises URINE_WINDOW_MISSING.
1. Lowest glucose	lowest_glucose	From DMS 5.3.A readings: minimum of values < 45 mg/dL. If none low → “Not Low”. If no readings → “Not Tested”.
2. Hypoglycaemia episodes	hypoglycemia_episodes	Count of DMS glucose readings < 45 mg/dL.
4. Highest glucose	highest_glucose	Maximum of DMS readings > 125 mg/dL. If none high → “Not High”. Insulin follow-up is asked when this is a number > 125.
6. Metabolic acidosis	metabolic_acidosis	Yes if any pH reading < 7.2. No if every recorded pH is ≥ 7.2. Empty if no numeric pH.

Quick copy of the weight and time building blocks

These sit under several forms. They are listed once here so the per-form tables stay short.

Screen field	Stored / API field	Formula
Grams → kilograms	birth_weight, growth_a.weight_g	kg = grams ÷ 1000.
GA as total days	gestation_weeks, gestation_days	days = weeks × 7 + days (0–6).
PMA date for a target week	used by Forms H, I, J, K	date = Day 1 Date (DOB) + (target weeks × 7 − GA days at birth).
PMA weeks+days on a given date	Forms F, G, K	PMA days = GA days at birth + (event date − DOB) in days. For DOL-based screens, postnatal days = DOL − 1. Then weeks = floor(/7), leftover days = remainder.
Effective weight for feeds and urine	Helpers 4 and 5	Use DMS 5.7.A kg on/before the day if it is ≥ birth kg; otherwise birth kg. Physiological weight loss must not inflate ml/kg.
Urine ml/kg/hr	Helper 5 windows and total	rate = volume_ml ÷ (effective kg × hours). 8am–2pm and 2pm–8pm = 6 h; 8pm–8am = 12 h; daily total uses only the hours that were actually recorded.

Source in code (for developers): frontend-app/src/utils/datetime.js, caffeineDose.js, gestation.js, dayOfLife.js, deathDetails.js, mmlHelperSync.js, data/intergrowthVeryPreterm.js; backend/value_ranges.py, bpd_suggestion.py, main.py _pma_target_date; tablet lib/utils/mml_helper_linkages.dart and matching form screens.